

SOFTWARE REQUIREMENTS SPECIFICATION

Synthetic Speech & Voice-Spoofing Detection System

Decision Support for Fraud Prevention through AI-Based Spoofed-Audio Detection

Lab 3 — Applying Requirements Engineering to Our Own Project

Merged Team Edition — combines and reconciles both members' independently revised Lab 3 drafts into one submission-ready document (see Section 7.3, Round 4).

Group Members

Donia Mabrouk

Iheb Zemzemi

Malak Ben Salem

AI Course — Semester Project, MUST University

September 2026 — Version 3.0 (Merged)

Table of Contents

1. Introduction

- 1.1 Purpose
- 1.2 Scope
- 1.3 Definitions, Acronyms, Abbreviations
- 1.4 References
- 1.5 Overview of the Document

2. Overall Description

- 2.1 Product Perspective
- 2.2 Product Functions
- 2.3 User Classes and Characteristics
- 2.4 Operating Environment
- 2.5 Design and Implementation Constraints
- 2.6 Assumptions and Dependencies

3. System Features / Use Cases

- 3.1 Actors
- 3.2 Use Case Diagram
- 3.3 Detailed Use Case Descriptions

4. Functional Requirements

5. Non-Functional Requirements

6. Requirements Traceability Matrix

7. Appendices

- 7.1 Elicitation Log
- 7.2 Requirements Analysis Table (MoSCoW)
- 7.3 Validation & Review Log
- 7.4 Open Parameters & Questions
- 7.5 Lab 3 Checkpoints & Instructor Sign-off
- 7.6 Glossary

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional requirements, non-functional (quality) requirements, and project/research constraints for the Synthetic Speech & Voice-Spoofing Detection System — an AI-based decision-support system that assesses short speech recordings as authentic, synthetic, or uncertain, with a confidence score and an operational risk level, to support fraud prevention. It consolidates and reconciles the elicitation, analysis, specification, use case modeling, and validation work independently produced by both team members for Lab 3 (Requirements Engineering) into a single reference for the project's design, implementation, and testing across the semester.

1.2 Scope

In scope: analysis of short recorded speech submitted through an analyst dashboard or a REST API; a three-way prediction (authentic / synthetic / uncertain) with a numeric confidence score and a categorical operational risk level (LOW / MEDIUM / HIGH); non-audio analysis metadata logging; an analyst history and annotation view; a shared evaluation harness comparing a classical baseline against the primary CNN detector; containerized deployment; and operational log review for the team's own operations/security administrator.

Research scope: the detector is trained, validated, tuned, and model-selected on FoR-2sec (Fake-or-Real, 2-second version; multi-engine TTS speech) only. The frozen final model and decision thresholds are then evaluated exactly once on DEEP-VOICE (RVC voice-conversion speech), a dataset never used for training, tuning, or model selection, to measure generalization from text-to-speech attacks to voice-conversion attacks. This external-test protocol, and the dataset-integrity rules that protect it, are treated as project/research constraints (Section 5, PC-xx) rather than runtime system behavior, since they govern how the team works, not what the deployed system does.

The system is a decision-support prototype, not a production-grade biometric authenticator, and does not claim to detect every present or future synthesis method.

Out of scope (Won't have this semester):

- Live real-time telephony or streaming detection — the system analyzes submitted recordings only.
- Defense against adversarial evasion attacks on the detector itself (documented as a limitation, not built).
- Fine-tuning of pretrained front-ends (wav2vec2 / WavLM) and spectrogram explainability (parking-lot work).
- A second external test dataset or alternative pretrained model families beyond the agreed CNN + classical baseline.
- Production multi-tenant identity management, downstream case-management or payment systems, and the choice of a commercial hosting vendor.

1.3 Definitions, Acronyms, Abbreviations

Term	Meaning
SRS	Software Requirements Specification
FR / NFR / PC	Functional Requirement / Non-Functional (quality) Requirement / Project or Research Constraint — process and methodology rules kept separate from runtime NFRs (Section 5)
MoSCoW	Prioritization scheme: Must have, Should have, Could have, Won't have
UC / UN	Use Case / User Need (elicited stakeholder need, Appendix 7.1)
Fraud Prevention Analyst	The human user who submits suspicious recorded audio through the dashboard and interprets the system's output as decision support
API Client	External software system that submits audio through the REST interface; distinct from the human analyst
Prediction	Exactly one classification label: authentic, synthetic, or uncertain
Confidence	Numeric model certainty on a 0.00–1.00 scale; not itself an operational risk level
Risk level	Operational priority category — LOW, MEDIUM, or HIGH — derived from prediction + confidence by a versioned mapping policy, distinct from Confidence
Uncertain	A normal, valid decision outcome used when configured decision thresholds do not justify an authentic or synthetic label — not an error state
Inference preprocessing profile	The documented preprocessing applied identically at inference: 16 kHz resampling, silence/voice-activity trimming, and the feature transformation the deployed model expects
FoR-2sec	Fake-or-Real dataset, 2-second version: real speech paired with synthetic speech from several TTS engines; the sole development dataset (training, validation, model selection, in-domain testing)
DEEP-VOICE	Dataset of real speech from public figures and RVC voice-conversion speech; used only once, as a frozen external generalization test — never for training, tuning, or model selection
RVC	Retrieval-based Voice Conversion
EER	Equal Error Rate — the operating point where false-acceptance and false-rejection rates are equal
ROC-AUC	Area Under the Receiver Operating Characteristic curve
MFCC	Mel-Frequency Cepstral Coefficients (the classical baseline model's features)
CNN	Convolutional Neural Network (the required primary deep model, over log-mel spectrograms)
ECE	Expected Calibration Error

1.4 References

- Project Proposal & Requirements Document — Synthetic Speech & Voice-Spoofing Detection for Fraud Prevention (internal team document, September 2026)
- Execution Reference — 14-week SDLC roadmap (internal team document)

- AI Course, Lab 2 and Lab 3 handouts (Requirements Engineering), MUST University
- IEEE 830 / ISO/IEC/IEEE 29148 — requirements specification structure (as introduced in Lab 2)
- Fake-or-Real (FoR) dataset documentation and its 2-second packaging
- DEEP-VOICE dataset and accompanying study, “Real-time Detection of AI-Generated Speech for DeepFake Voice Conversion”
- In-class peer review notes on the first Lab 3 draft (see Section 7.3)

1.5 Overview of the Document

Section 2 describes the product at a high level. Section 3 presents the merged use case model. Section 4 gives the testable functional requirements; Section 5 gives runtime non-functional requirements and, in a clearly separated subsection, the project/research constraints that govern methodology rather than system behavior. Section 6 traces every requirement to its source and use case. Section 7 contains the Lab 3 working artifacts: elicitation log, requirements analysis table, a four-round validation and review log (internal merge, in-class peer review, expert review, and this document's own team merge), open parameters, checkpoint sign-offs, and a glossary.

2. Overall Description

2.1 Product Perspective

The system is a new, standalone academic project. It is designed as an add-on fraud-detection control that an organization such as a bank or call center could place alongside its existing systems: an external call-recording or audio-capture system supplies recordings (via the API Client role), and flagged results can be passed on to the fraud team's own review process. Internally, it separates an offline training/evaluation pipeline (datasets, baseline and CNN training, generalization experiments) from an online inference service (REST API, risk and confidence layer, dashboard) and a lightweight, privacy-conscious logging layer that operations staff can inspect independently of analysis metadata.

2.2 Product Functions

- Audio submission through the analyst dashboard and through a REST API
- Input validation against documented format, size, and duration limits
- Preprocessing under a versioned inference preprocessing profile (resampling, trimming, feature extraction)
- Classification into authentic / synthetic / uncertain, with a confidence score and a LOW/MEDIUM/HIGH risk level
- Non-audio analysis metadata logging and an analyst history view
- Optional analyst annotation of a result, access-restricted to authenticated analysts
- Model evaluation: classical baseline vs. CNN through one shared evaluation harness, with full metric reporting

- External generalization test: the frozen FoR-2sec-tuned model evaluated exactly once on DEEP-VOICE
- Evaluation reporting for compliance/research review
- Operational service-log review for the operations/security administrator, separate from analysis metadata
- Containerized deployment of the API and dashboard

2.3 User Classes and Characteristics

User Class	Characteristics
Fraud Prevention Analyst	Primary user. Reviews suspicious recordings daily; needs a clear, fast result — prediction, confidence, and risk level — rather than raw model output. Limited ML background; a false accusation is a serious concern.
API Client	External system (configured by a technical integration owner) that submits recordings through the API; cares about a stable request/response contract, documented limits, and predictable errors, consistent with the dashboard's own logic.
Compliance / Security Officer	Occasional user. Needs data minimization, a configurable retention policy, safe (non-raw-audio) logging, and evaluation results — including generalization performance — to judge whether the detector can be trusted.
Operations / Security Administrator	Technical operator. Reviews service/operational logs and monitoring, confirms model/config version visibility, and checks for safe failure behavior and no internal-data leakage — distinct from the Compliance Officer's audit concerns.
Call-center / Bank Operations Manager	Occasional business stakeholder. Cares about workflow fit, acceptable false-positive/false-negative trade-offs, measurable latency, and incident review; not a direct system operator.
ML / Research Engineer (internal)	Trains and evaluates the models, maintains the evaluation harness, and is the only role that may run the external generalization test.
Caller / Customer (indirect)	Person whose voice is analyzed. Never uses the system directly but is affected by recording retention and by false “synthetic” results; expects human review before any high-impact decision.
Course Instructor	Reviews SDLC artifacts, traceability, and the final demo against the rubric.

2.4 Operating Environment

Training and experiments run on Google Colab and Kaggle Notebooks (GPU-backed) with datasets on a shared Google Drive; experiment runs are tracked in Weights & Biases. The inference API and dashboard run as Docker containers. Users need only a modern web browser (dashboard) or a standard HTTP client (API); operations staff use the same containerized deployment's logs/monitoring surface.

2.5 Design and Implementation Constraints

- One primary deep model: the deployed detector is a single CNN over log-mel spectrograms (PC-08). A classical MFCC-based model is built only as an evaluation baseline, never as a deployment candidate.

- Metrics: accuracy, precision, recall, F1, EER, ROC-AUC, and the confusion matrix are required for every evaluation (PC-06). t-DCF is excluded because it is defined around the ASVspoof protocol, which this project does not use.
- Scope discipline: exactly two stretch goals are permitted — a robustness/degradation curve and confidence calibration — so they cannot displace the core MVP.
- Timeline: one 14-week semester, three team members.
- All dataset-integrity, model-freeze, and external-test protocol rules are treated as Section 5 project/research constraints (PC-xx), not as runtime NFRs, so that process discipline is not mistaken for a system quality attribute.

2.6 Assumptions and Dependencies

- Assumption: the FoR-2sec packaging preserves enough of the original release's TTS-engine diversity. Dependency: the Week 2 dataset audit (PC-01) confirms or documents otherwise.
- Assumption: DEEP-VOICE, being RVC-only with a real class imbalance between its bonafide and spoof clips, is still a meaningful external test provided that imbalance is stated with every reported result (NFR-- see Section 5.2); exact class counts are confirmed against the dataset files directly rather than assumed here.
- Dependency: GPU compute through Colab (possibly Pro, shared) or Kaggle's free tier (~30 GPU-hours/week per account).
- Dependency: the Weights & Biases free tier is sufficient for shared experiment tracking.
- Assumption: all three members remain available for the full roadmap, and instructor checkpoints do not block the next phase.

3. System Features / Use Cases

3.1 Actors

Actor	Type	Goals with the system
Fraud Prevention Analyst	Primary, human	Submit recordings via the dashboard, read results, review analysis history, annotate results
API Client	Primary, external system	Submit recordings through the REST API and receive a structured result
ML / Research Engineer	Primary, human (internal)	Evaluate the models (baseline vs. CNN) and run the one-time external generalization test
Compliance Officer	Secondary, human	Read evaluation results and logs to judge trustworthiness and auditability
Operations / Security Administrator	Secondary, human	Review service/operational logs and monitoring; confirm safe, observable operation

3.2 Use Case Diagram

Each use case bubble is a goal of an actor, except two shared, non-actor-facing use cases reused by more than one base use case: Validate & Analyze Audio (the common validation, preprocessing, detection, and risk-scoring pipeline shared by both submission channels) and Record Analysis Metadata (the common logging step triggered by a successful analysis or by an annotation). Neither has an independent actor and neither is included by anything else, so this model avoids the chained-include problem raised in the team's peer review (Section 7.3). Annotate Result is optional, so it extends Review Analysis History rather than being included by it.

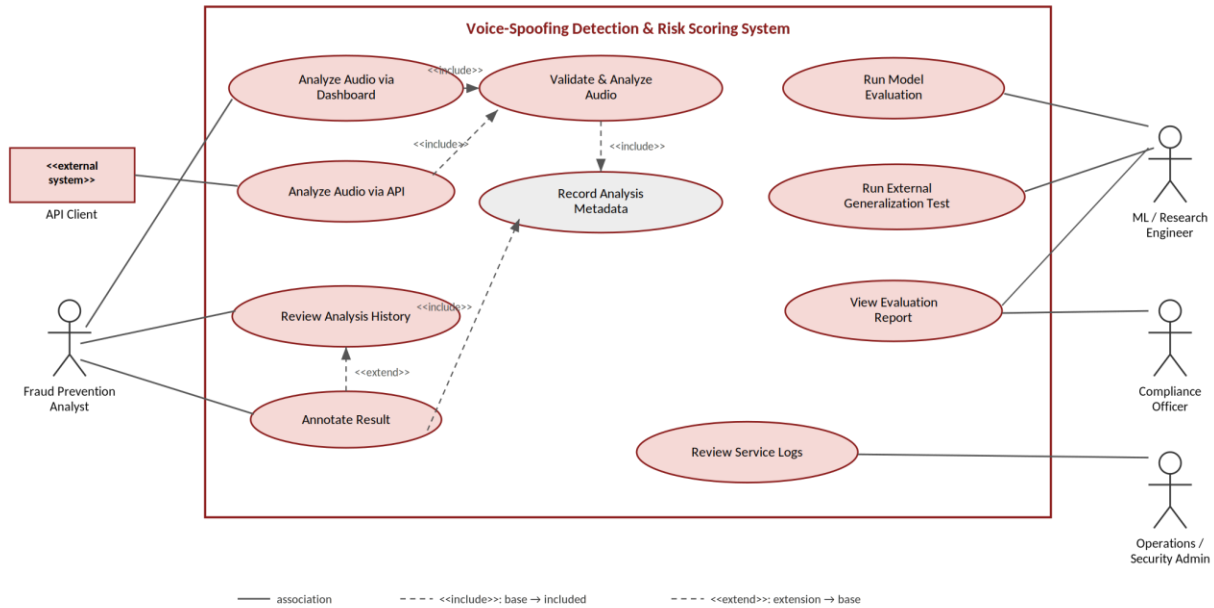


Figure 1 — Merged use case model: submission, review, research evaluation, and operations.

3.3 Detailed Use Case Descriptions

Ten use cases are documented below: eight actor-facing goals and two shared, included support use cases. This set merges both members' independent revisions — the split dashboard/API use cases and the Operations actor (originally modeled by one member) with the research-evaluation use cases and analyst annotation feature (originally modeled by the other), so that every functional requirement in Section 4 traces to a real use case rather than a diagram-only box.

UC-01 — Analyze Audio via Dashboard

Field	Description
Use Case ID	UC-01
Name	Analyze Audio via Dashboard
Primary Actor	Fraud Prevention Analyst
Preconditions	Dashboard is available; the input policy (formats/size/duration limits) is published; the deployed model/config is loaded.

Field	Description
Main Flow	1. The analyst selects an audio file. 2. The system validates and analyzes the file (include: Validate & Analyze Audio, FR-03–FR-07). 3. The dashboard displays the prediction, confidence, risk level, and a decision-support notice (FR-08).
Alternate Flows	A1. Unsupported, corrupt, or policy-violating file: the system returns a structured validation error; no inference and no successful result record are created (NFR-06). A2. Detector or service failure: the system returns a structured service error; it does not fabricate a prediction (NFR-06). Note: “uncertain” is a normal successful outcome, not an exception.
Postconditions	A successful result is displayed and traceable by request ID; raw-audio retention follows NFR-03.
Linked Requirements	FR-01, FR-03–FR-09, NFR-01–NFR-07, NFR-09

UC-02 — Analyze Audio via API

Field	Description
Use Case ID	UC-02
Name	Analyze Audio via API
Primary Actor	API Client
Preconditions	The API service is reachable; the input policy is published; the deployed model/config is loaded.
Main Flow	1. The API Client sends one audio file to POST /analyze-audio (FR-02). 2. The system validates and analyzes the file (include: Validate & Analyze Audio, FR-03–FR-07). 3. The system returns structured JSON containing the prediction, confidence, risk level, and request ID (FR-05–FR-07).
Alternate Flows	A1. Invalid or unsupported input: the system returns a structured client error identifying the violated policy; no inference result is created. A2. Inference or service failure: the system returns a structured server error without an internal stack trace or a fabricated prediction (NFR-02, NFR-06).
Postconditions	The client receives either one valid structured analysis response or one structured error response; successful analyses are traceable by request ID.
Linked Requirements	FR-02–FR-07, FR-09, NFR-02–NFR-04, NFR-06–NFR-09

UC-03 — Validate & Analyze Audio (shared, included)

Field	Description
Use Case ID	UC-03
Name	Validate & Analyze Audio
Included by	UC-01 (Analyze Audio via Dashboard), UC-02 (Analyze Audio via API)

Field	Description
Preconditions	Called only from UC-01 or UC-02; a submitted file reference is available.
Main Flow	1. The system validates the file's readability, format, size, and duration against the documented limits (FR-03). 2. The system applies the versioned inference preprocessing profile (FR-04). 3. The detector produces exactly one prediction — authentic, synthetic, or uncertain (FR-05). 4. The system computes a confidence score in [0.00, 1.00] (FR-06). 5. The system derives a LOW/MEDIUM/HIGH operational risk level from the versioned risk-mapping policy (FR-07). 6. The system triggers metadata logging for the successful result (include: Record Analysis Metadata, FR-09).
Alternate Flow	1a. Validation fails: return to the calling use case's alternate flow A1 without proceeding to steps 2–6.
Postconditions	A prediction, confidence, and risk level are available to the calling use case, and the result is logged.
Linked Requirements	FR-03, FR-04, FR-05, FR-06, FR-07

UC-04 — Record Analysis Metadata (shared, included)

Field	Description
Use Case ID	UC-04
Name	Record Analysis Metadata
Included by	UC-03 (Validate & Analyze Audio), UC-06 (Annotate Result)
Preconditions	A successful analysis or a new annotation exists to record.
Main Flow	1. The system writes a metadata record containing request ID, timestamp, model/config version, prediction, confidence, risk level, and latency (FR-09). 2. No raw audio is included in the record (NFR-04).
Alternate Flow	None — this use case only runs after its caller has already succeeded.
Postconditions	The record is stored and retrievable by request ID or timestamp, and is traceable per NFR-07.
Linked Requirements	FR-09, NFR-04, NFR-07

UC-05 — Review Analysis History (extended by: Annotate Result)

Field	Description
Use Case ID	UC-05
Name	Review Analysis History
Actor	Fraud Prevention Analyst
Preconditions	The dashboard is available; zero or more prior successful analyses may exist.

Field	Description
Main Flow	1. The analyst opens the analysis history. 2. The system retrieves stored non-audio metadata and marks records labeled synthetic or uncertain as flagged (FR-09, FR-10). 3. The system displays request ID/time, prediction, confidence, risk level, and model/config version for each available record. 4. The analyst reviews past results as decision-support records only. Extension point (after step 3): the analyst may annotate a result (Annotate Result, FR-11).
Alternate Flows	A1. No prior records: display an empty-history state. A2. History store unavailable: display a clear retrieval error without exposing internal details.
Postconditions	No analysis result is modified by viewing; displayed records remain traceable to their model/config version.
Linked Requirements	FR-09, FR-10, NFR-02, NFR-04, NFR-05, NFR-07

UC-06 — Annotate Result (extends: Review Analysis History)

Field	Description
Use Case ID	UC-06
Name	Annotate Result
Actor	Fraud Prevention Analyst (authenticated)
Preconditions	The analyst is signed in and authenticated; a result from UC-05 is open.
Main Flow	1. The analyst attaches a note to the open result. 2. The system verifies the requester is an authenticated fraud-analyst account (FR-11). 3. The system stores the annotation next to the unchanged original result. 4. The system logs the annotation event (include: Record Analysis Metadata, UC-04).
Alternate Flow	2a. The requester is unauthenticated or not an analyst: the system denies the request; no annotation is stored (FR-11).
Postconditions	The annotation is stored and auditable; the original prediction, confidence, and risk level are unchanged.
Linked Requirements	FR-11

UC-07 — Run Model Evaluation

Field	Description
Use Case ID	UC-07
Name	Run Model Evaluation
Actor	ML / Research Engineer
Preconditions	The FoR-2sec dataset audit is documented (PC-01); the group-aware split exists or its limitation is documented (PC-02); the classical baseline and the CNN are both trained.

Field	Description
Main Flow	1. The engineer selects the baseline (MFCC-based) and the CNN model to evaluate. 2. The system runs both models through the same evaluation harness on the same FoR-2sec test split (FR-12). 3. The harness computes accuracy, precision, recall, F1, EER, ROC-AUC, and a confusion matrix that includes the uncertain outcome, for both models (FR-13). 4. The report identifies the dataset/split and states class counts/distribution alongside the metrics (FR-14). 5. The engineer may optionally run the two agreed stretch-goal analyses on the CNN: a robustness/degradation curve (FR-17) and confidence calibration with a reliability diagram and ECE (FR-18).
Alternate Flow	1a. The FoR-2sec split is not group-aware and no documented limitation exists: the evaluation must not be reported until PC-02 is satisfied or the limitation is recorded.
Postconditions	In-domain baseline-vs-CNN results (and any stretch-goal results) exist and are available to View Evaluation Report (UC-09).
Linked Requirements	FR-12, FR-13, FR-14, FR-17, FR-18

UC-08 — Run External Generalization Test

Field	Description
Use Case ID	UC-08
Name	Run External Generalization Test
Actor	ML / Research Engineer
Preconditions	The dataset audit is documented (PC-01); model selection and threshold tuning on FoR-2sec are finished and the final model is frozen and recorded (PC-03, PC-04); the shared evaluation harness is available.
Main Flow	1. The engineer selects the frozen final model and DEEP-VOICE as the evaluation set. 2. The system runs the model on DEEP-VOICE with no training, tuning, or threshold change (PC-05, FR-15). 3. The evaluation harness computes the required metrics (FR-13). 4. The system records the run configuration and results so the run can be reproduced (PC-07). 5. The engineer reports the DEEP-VOICE results next to the FoR-2sec in-domain results, with the gap quantified and the RVC-only scope and class imbalance stated (FR-16).
Alternate Flow	1a. The model is not frozen, or the audit is missing: the test must not be run; the engineer completes the missing step first (PC-04).
Postconditions	One reproducible external-test result exists and is available to View Evaluation Report (UC-09). The result is never used to retrain or retune (PC-05).
Linked Requirements	FR-13, FR-15, FR-16

UC-09 — View Evaluation Report

Field	Description
Use Case ID	UC-09
Name	View Evaluation Report
Actor	Compliance Officer (primary); ML / Research Engineer (secondary)
Preconditions	At least one Run Model Evaluation (UC-07) result exists.
Main Flow	1. The actor requests the evaluation report. 2. The system assembles the in-domain baseline-vs-CNN results (from UC-07) into one report (FR-13, FR-14). 3. If an external generalization result (UC-08) exists, the system includes it as a distinct, clearly labeled section (FR-16); if not, that section is marked “not yet available” rather than omitted (FR-19). 4. The system displays the report and offers an export option.
Alternate Flow	2a. No Run Model Evaluation result exists yet: the system shows an empty-state message instead of a report.
Postconditions	The report is available for review/export; no analysis or evaluation state is changed.
Linked Requirements	FR-13, FR-14, FR-16, FR-19

UC-10 — Review Service Logs

Field	Description
Use Case ID	UC-10
Name	Review Service Logs
Actor	Operations / Security Administrator
Preconditions	The administrator has operational access to the deployed containers/monitoring surface.
Main Flow	1. The administrator opens the service/operational log or monitoring view. 2. The system presents logs showing model/config version, request volumes, latency, and error rates, with no raw audio and no internal secrets (NFR-02, NFR-04). 3. The administrator confirms safe, observable operation or identifies an incident for follow-up.
Alternate Flow	1a. Monitoring surface unavailable: the administrator is shown a clear connectivity error rather than stale or fabricated data.
Postconditions	No analysis or evaluation state is changed; the review itself may be logged for audit.
Linked Requirements	FR-20, NFR-02, NFR-04, NFR-07

4. Functional Requirements

Each requirement uses “shall”, is traceable to a source (Section 6), and has an acceptance criterion. Operational values that the source documents do not fix (maximum upload size, latency target,

retention period, etc.) are referenced as named parameters, defined and tracked to closure in Appendix 7.4, rather than invented here.

4.1 Audio Submission & Validation

ID	Requirement	Priority	Acceptance Criteria
FR-01	The system shall allow a Fraud Prevention Analyst to submit one supported short audio file for analysis through the dashboard.	Must	An analyst can upload a file meeting the documented policy on the dashboard and an analysis starts.
FR-02	The system shall allow an API Client to submit one supported short audio file for analysis through POST /analyze-audio.	Must	A documented POST request with a supported file returns a request-specific structured response.
FR-03	The system shall validate each submitted file against readability, SUPPORTED_AUDIO_FORMATS, MAX_UPLOAD_MB, and MAX_AUDIO_SECONDS before inference.	Must	An unsupported, corrupt, or limit-violating file is rejected before inference with a structured reason.

4.2 Preprocessing, Detection & Risk

ID	Requirement	Priority	Acceptance Criteria
FR-04	The system shall transform each accepted file using the versioned inference preprocessing profile before model inference.	Must	A known input processed under a selected profile version produces the expected model-ready representation.
FR-05	The system shall return exactly one prediction label from {authentic, synthetic, uncertain} for each successful analysis.	Must	Every completed analysis returns exactly one of the three labels.
FR-06	The system shall return a confidence value in the inclusive range 0.00 to 1.00 for each successful analysis.	Must	Every successful result contains a numeric confidence in [0.00, 1.00].
FR-07	The system shall return one operational risk level from {LOW, MEDIUM, HIGH} using the versioned risk-mapping policy associated with the deployed model.	Should	Every successful result contains exactly one risk level from the versioned policy; changing the policy version changes future mappings.

4.3 Dashboard, Logging & History

ID	Requirement	Priority	Acceptance Criteria
FR-08	The dashboard shall display the prediction, confidence, risk level, and a decision-support notice together for each successful analysis.	Must	After a submission, all four elements appear on screen for that analysis.
FR-09	The system shall record analysis metadata — request ID, timestamp, model/config version, prediction, confidence, risk level, and latency — for each successful analysis, without raw audio.	Must	After N analyses the log holds N entries with all seven fields, retrievable by request ID or timestamp; no entry contains raw audio.

ID	Requirement	Priority	Acceptance Criteria
FR-10	The dashboard shall allow a Fraud Prevention Analyst to view stored analysis-history metadata, with results labeled synthetic or uncertain marked as flagged.	Should	The history view lists records with all stored fields, most recent first, and every synthetic/uncertain result carries a flagged marker; an empty state is shown when no records exist.
FR-11	The dashboard shall allow an authenticated Fraud Prevention Analyst to attach an annotation to a result without changing the original result. An unauthenticated or non-analyst request to annotate shall be denied.	Could	A saved annotation appears next to the original result, whose prediction and confidence are unchanged. An unauthenticated or non-analyst annotation request is denied.

4.4 Model Evaluation & Reporting

ID	Requirement	Priority	Acceptance Criteria
FR-12	The evaluation subsystem shall evaluate the classical MFCC-based baseline and the CNN through the same evaluation harness and test split.	Must	Both models' results are produced by the same metrics module on the same FoR-2sec test split.
FR-13	The evaluation subsystem shall compute accuracy, precision, recall, F1, EER, ROC-AUC, and a confusion matrix that includes the uncertain outcome, for a labeled evaluation set.	Must	Every evaluation output contains all seven items.
FR-14	The evaluation report shall identify the evaluated dataset and split, and state class counts/distribution alongside the reported metrics.	Must	Every report names its dataset/split and states class counts before or alongside the metrics table.
FR-15	The evaluation subsystem shall evaluate the frozen final model on DEEP-VOICE exactly once and report the results as a separate external-generalization section (see PC-04, PC-05).	Must	The report has a distinct DEEP-VOICE section whose run references the frozen model checkpoint, and only one such run is ever reported.
FR-16	Every reported external-test result shall be stated next to the in-domain FoR-2sec result, with the gap quantified and DEEP-VOICE's RVC-only scope and class imbalance stated.	Must	Review of the evaluation report finds the in-domain comparison, the quantified gap, and both caveats in the external-generalization section.
FR-17	The evaluation subsystem shall measure EER at increasing levels of controlled noise, compression, and bandwidth reduction.	Could	A degradation curve (clean → mild → moderate → severe) is produced from the FoR-2sec test split.
FR-18	The evaluation subsystem shall calibrate the model's confidence and assess it with a reliability diagram and ECE.	Could	Reliability diagram and ECE are reported before and after calibration.
FR-19	The system shall produce an evaluation report on request and shall mark the external-generalization section as "not yet available" if FR-15 has not been run.	Should	Requesting the report before and after the external test shows the placeholder, then the results.

4.5 Operations

ID	Requirement	Priority	Acceptance Criteria
FR-20	The system shall expose service/operational logs — model/config version, request volume, latency, and error rate — for review by the Operations / Security Administrator, separate from analysis metadata and containing no raw audio.	Should	The administrator can open a log/monitoring view showing all four elements with no raw audio or analysis-specific personal data present.

5. Non-Functional Requirements

These are runtime quality attributes of the deployed system. Rules that govern how the team works with data and models — not what the running system does — are kept separate as Project/Research Constraints (Section 5.7), per the peer-review correction recorded in Section 7.3.

5.1 Performance

ID	Requirement	Priority	Acceptance Criteria
NFR-01	The system shall complete at least 95% of valid analysis requests within L_MAX seconds under the documented reference workload.	Should	A load/latency test after L_MAX is baselined (Appendix 7.4) shows $\geq 95\%$ of requests completing within L_MAX.

5.2 Security

ID	Requirement	Priority	Acceptance Criteria
NFR-02	The system shall not expose model weights, training-data locations, secrets, or internal stack traces in dashboard or API responses.	Must	Response and security inspection finds none of the four categories in any response, including error responses.

5.3 Privacy

ID	Requirement	Priority	Acceptance Criteria
NFR-03	The system shall retain uploaded raw audio for no longer than RETENTION_HOURS and shall remove it when that period expires.	Must	A retention test using a shortened test value confirms the audio file is absent from storage after the period, with the metadata record intact.
NFR-04	The system shall exclude raw uploaded audio and direct personal identifiers from normal application and operational logs.	Must	Log inspection across analysis metadata and service logs finds no raw audio and no direct personal identifiers.

5.4 Usability & Decision Support

ID	Requirement	Priority	Acceptance Criteria
NFR-05	The user interface shall identify all analysis outputs as decision support and shall not present an uncertain output as a definitive fraud finding.	Must	UX content review finds decision-support framing throughout and no wording such as “confirmed fake”.

5.5 Reliability & Observability

ID	Requirement	Priority	Acceptance Criteria
NFR-06	The system shall return a structured error for invalid input or inference failure without creating a normal successful analysis result.	Must	Fault-injection and negative-input tests confirm a structured error and no fabricated success record.
NFR-07	The system shall make each successful analysis traceable by request ID, model/config version, timestamp, and measured latency.	Must	A stored record for any successful analysis contains all four traceability fields.

5.6 Compatibility, Deployability & Maintainability

ID	Requirement	Priority	Acceptance Criteria
NFR-08	The dashboard and API shall enforce the same documented SUPPORTED_AUDIO_FORMATS and input-limit policy.	Should	Cross-interface conformance tests confirm identical accept/reject behavior for the same file across both interfaces.
NFR-09	The deployed detector shall load a model version together with its matching preprocessing profile and risk-threshold configuration version.	Should	Deployment/configuration inspection confirms all three versions are bound and logged together.
NFR-10	The API and dashboard shall be deployable together as containers using only the README instructions.	Must	On a fresh machine with Docker, following the README starts both services and a test submission succeeds.
NFR-11	The project shall include automated tests for preprocessing, the risk and confidence layer, and the API end to end, run in continuous integration.	Should	The CI run executes tests for all three parts and passes on the final release.

5.7 Project / Research Constraints (PC) — not runtime NFRs

These govern the team's dataset handling, model development, and evaluation methodology. They are listed separately from Sections 5.1–5.6 because they describe process discipline the team must follow, not a quality attribute of the running system — conflating the two was a specific issue raised in peer review (Section 7.3).

ID	Constraint	Priority	Source
PC-01	The FoR-2sec / DEEP-VOICE dataset audit (provenance, speakers/sources, duplicates, class balance, licenses) shall be completed and documented before any model training.	Must	Execution Phase 2; Proposal §6.1
PC-02	The FoR-2sec train/validation/test split shall be group-aware, so no speaker or source recording appears in more than one split, wherever the metadata supports it; otherwise the limitation shall be documented.	Must	Proposal §6.1
PC-03	All model selection, hyperparameter tuning, and threshold tuning shall use FoR-2sec only.	Must	Proposal §6.1/7.1
PC-04	The final selected model and its decision thresholds shall be frozen before DEEP-VOICE is evaluated.	Must	Proposal §6.1; Execution Phase 4

ID	Constraint	Priority	Source
PC-05	DEEP-VOICE shall be evaluated exactly once, as the frozen external test; its results shall not trigger retraining, retuning, or threshold adjustment.	Must	Proposal §6.1; Execution standing rules
PC-06	The required evaluation metrics are accuracy, precision, recall, F1, EER, ROC-AUC, and the confusion matrix; t-DCF is excluded from the current methodology.	Must	Proposal §8; Execution Phase 3
PC-07	Each training and evaluation run shall retain its dataset/config/code-version information with its results so it can be reproduced.	Must	Execution Phase 0 / standing rules
PC-08	The required primary deep model is a CNN over log-mel spectrograms; additional architectures are parking-lot work and shall not displace the MVP.	Should	Proposal §7.1–7.2
PC-09	Exactly two stretch goals are permitted for this iteration: a robustness/degradation curve (FR-17) and confidence calibration (FR-18).	Should	Proposal §7.2

6. Requirements Traceability Matrix

Maps each requirement (including PC-xx constraints) to its source and, for runtime FR/NFR items, its use case. Sources combine role-play elicitation notes (UN-xx, Appendix 7.1), document analysis (DOC-xx, Appendix 7.1), and in-class peer-review findings where applicable. Test-case IDs are assigned in the Testing phase (Week 13).

ID	Source	Use Case	Test Case
FR-01	UN-01; DOC-02 (Proposal §5)	UC-01	TBD (Week 13)
FR-02	UN-05; DOC-02 (Proposal §5)	UC-02	TBD (Week 13)
FR-03	UN-01, UN-05; peer review	UC-03	TBD (Week 13)
FR-04	DOC-07 (Execution Phase 2)	UC-03	TBD (Week 13)
FR-05	UN-01; DOC-01 (Proposal §4)	UC-03	TBD (Week 13)
FR-06	UN-01; DOC-02 (Proposal §5)	UC-03	TBD (Week 13)
FR-07	UN-01; peer review (risk-level elaboration)	UC-03	TBD (Week 13)
FR-08	UN-01; DOC-01 (Proposal §4–5)	UC-01	TBD (Week 13)
FR-09	UN-03; DOC-08 (Execution Phase 7)	UC-03, UC-04 (include)	TBD (Week 13)
FR-10	UN-01	UC-05	TBD (Week 13)
FR-11	UN-02	UC-06 (extend of UC-05)	TBD (Week 13)
FR-12	UN-04; DOC-01 (Proposal §4.1)	UC-07	TBD (Week 13)
FR-13	UN-04; DOC-04 (Proposal §8)	UC-07, UC-08, UC-09	TBD (Week 13)

ID	Source	Use Case	Test Case
FR-14	UN-04; DOC-03 (Proposal §6.1/8)	UC-07, UC-09	TBD (Week 13)
FR-15	UN-03; DOC-05 (Proposal §7.1)	UC-08	TBD (Week 13)
FR-16	UN-04; DOC-05 (Proposal §7.1)	UC-08, UC-09	TBD (Week 13)
FR-17	DOC-06 (Proposal §7.2)	UC-07 (stretch-goal step)	TBD (Week 13)
FR-18	DOC-06 (Proposal §7.2)	UC-07 (stretch-goal step)	TBD (Week 13)
FR-19	UN-03	UC-09	TBD (Week 13)
FR-20	UN-06	UC-10	TBD (Week 13)
NFR-01	UN-04	UC-01, UC-02	TBD (Week 13)
NFR-02	UN-03, UN-06; threat model	UC-02, UC-10	TBD (Week 13)
NFR-03	UN-02, UN-03	System-wide (data lifecycle)	TBD (Week 13)
NFR-04	UN-03, UN-06; DOC-08	UC-04, UC-10	TBD (Week 13)
NFR-05	UN-01, UN-02; DOC-01 (Proposal §3–4)	UC-01, UC-05	TBD (Week 13)
NFR-06	UN-01, UN-06; peer review	UC-01, UC-02	TBD (Week 13)
NFR-07	UN-03, UN-06	UC-04	TBD (Week 13)
NFR-08	UN-05	UC-01, UC-02	TBD (Week 13)
NFR-09	Execution reproducibility baseline	System-wide	TBD (Week 13)
NFR-10	UN-05; DOC-05 (Proposal §7.1)	System-wide	TBD (Week 13)
NFR-11	UN-04; DOC-08 (Execution Phase 10)	System-wide	TBD (Week 13)
PC-01	DOC-07 (Execution Phase 2); Proposal §6.1	— (process constraint)	—
PC-02	DOC-03 (Proposal §6.1)	— (process constraint)	—
PC-03	DOC-03 (Proposal §6.1/7.1)	— (process constraint)	—
PC-04	DOC-03; Execution Phase 4	— (process constraint)	—
PC-05	DOC-03; Execution standing rules	— (process constraint)	—
PC-06	DOC-04 (Proposal §8); Execution Phase 3	— (process constraint)	—
PC-07	DOC-07 (Execution Phase 0)	— (process constraint)	—
PC-08	DOC-06 (Proposal §7.1–7.2)	— (process constraint)	—
PC-09	DOC-06 (Proposal §7.2)	— (process constraint)	—

7. Appendices

7.1 Elicitation Log

Collected through role-play interviews inside the team (one member played the stakeholder, another interviewed) and through analysis of the project documents. These are prepared role-play artifacts, not interviews with real organizations — explicitly permitted by the Lab 3 handout. Six stakeholder personas (UN-01–UN-06) merge both members' independent elicitation passes; document-analysis notes are broken into DOC-01 through DOC-08, one per project-document section cited in Section 6.

ID	Technique	Participant / Source	Key Findings (raw notes)
UN-01	Role-play interview	Fraud Prevention Analyst	"Need a result quickly while reviewing a suspicious call." "Do not force a yes/no when the model is unsure." "Show confidence and a simple risk category." "I need previous cases/history." "If the file is bad, tell me what is wrong instead of failing silently."
UN-02	Role-play interview	Caller / Customer	"My voice recording is sensitive." "Do not keep the audio forever." "A false accusation is serious." "The model should assist a person, not automatically accuse or block me."
UN-03	Role-play interview	Security & Compliance Officer	"Need a configurable retention rule." "Logs should avoid raw audio." "Keep enough metadata to audit what happened — time, model version, result." "Do not expose training data, model internals, or stack traces."
UN-04	Role-play interview	Call-center / Bank Operations Manager	"The tool cannot add a huge delay to the workflow." "We need to know false positives and false negatives, not only accuracy." "If the service is unavailable, the analyst still needs a clear failure message." "External-test class imbalance must be stated with the metrics."
UN-05	Role-play interview	API Client / Integration Owner	"Need one stable endpoint and structured JSON." "Dashboard and API should agree on the same prediction and risk logic." "Document supported formats and size limits." "Give predictable client errors for bad uploads."
UN-06	Role-play interview	Operations / Security Administrator	"Need model/config version in logs." "No raw audio in normal logs." "Service should be containerized and observable." "Do not leak internal stack traces to clients."
DOC-01	Document analysis	Project Proposal §4 / §4.1	Detection is a three-way label (authentic/synthetic/uncertain), decision-support framing, not a definitive fraud finding; evaluation compares the CNN against a classical baseline on the same harness.
DOC-02	Document analysis	Project Proposal §5	System design: dashboard upload + REST endpoint (POST /analyze-audio) share one preprocessing pipeline; confidence reported per result.
DOC-03	Document analysis	Project Proposal §6.1	Dataset integrity rules: DEEP-VOICE never used for tuning; FoR-2sec split must be group-aware by

ID	Technique	Participant / Source	Key Findings (raw notes)
			speaker/source; no results reported before the dataset audit; freeze-before-external-test rule.
DOC-04	Document analysis	Project Proposal §8	Required evaluation metrics: accuracy, precision/recall/F1, EER, ROC-AUC, confusion matrix; t-DCF excluded (ASVspoof-specific, not used).
DOC-05	Document analysis	Project Proposal §7.1	External generalization protocol: freeze the FoR-2sec-tuned model, then evaluate once on DEEP-VOICE; report the gap and dataset caveats alongside the in-domain result.
DOC-06	Document analysis	Project Proposal §7.1–7.2	Primary model is a CNN over log-mel spectrograms; two agreed stretch goals: a robustness/degradation curve and confidence calibration (reliability diagram + ECE).
DOC-07	Document analysis	Execution Plan, Phase 0 / Phase 2	Reproducibility infrastructure: every run records environment, config, seed, dataset reference, checkpoint; Phase 2 defines the inference preprocessing profile.
DOC-08	Document analysis	Execution Plan, Phase 7 / Phase 10	Logging design (fields, no raw audio) and CI testing phase: automated tests for preprocessing, the risk/confidence layer, and the API run in CI before the final release.

7.2 Requirements Analysis Table (MoSCoW)

Lab 3 Step 2 output: raw needs turned into candidates, classified FR/NFR/PC, categorized, and prioritized. Won't-have items are kept so the scope boundary is explicit.

ID	Type	Category	Candidate (short)	Priority	Source
FR-01	FR	Submission	Analyst uploads audio on dashboard	Must	UN-01
FR-02	FR	Submission	API client submits via REST	Must	UN-05
FR-03	FR	Validation	Reject invalid files with reason	Must	UN-01, UN-05
FR-04	FR	Preprocessing	Versioned preprocessing profile	Must	DOC-07
FR-05	FR	Detection	Three-way prediction	Must	UN-01, DOC-01
FR-06	FR	Confidence	Numeric confidence [0,1]	Must	UN-01
FR-07	FR	Risk	LOW/MEDIUM/HIGH risk level	Should	UN-01
FR-08	FR	Dashboard	Show prediction/confidence/risk	Must	UN-01
FR-09	FR	Logging	Log every analysis, no raw audio	Must	UN-03
FR-10	FR	History	History view + flagged marker	Should	UN-01

ID	Type	Category	Candidate (short)	Priority	Source
FR-11	FR	Dashboard	Analyst annotation (incl. access control)	Could	UN-02
FR-12	FR	Evaluation	Baseline vs. CNN, same harness	Must	UN-04, DOC-01
FR-13	FR	Evaluation	Full metric set	Must	DOC-04
FR-14	FR	Evaluation reporting	Dataset/split + class distribution stated	Must	DOC-03
FR-15	FR	Generalization	Frozen model on DEEP-VOICE, once	Must	DOC-05
FR-16	FR	Generalization reporting	Gap + caveats stated with result	Must	UN-04, DOC-05
FR-17	FR	Robustness	Degradation curve (stretch)	Could	DOC-06
FR-18	FR	Calibration	Reliability diagram / ECE (stretch)	Could	DOC-06
FR-19	FR	Reporting	Report placeholder if not yet run	Should	UN-03
FR-20	FR	Operations	Service-log review for admin	Should	UN-06
NFR-01	NFR	Performance	$\leq L_MAX$ per clip (parameterized)	Should	UN-04
NFR-02	NFR	Security	No secrets/weights/stack traces exposed	Must	UN-03, UN-06
NFR-03	NFR	Privacy	Audio retention $\leq$ RETENTION_HOURS	Must	UN-02, UN-03
NFR-04	NFR	Privacy	No raw audio / PII in normal logs	Must	UN-03, UN-06
NFR-05	NFR	Decision support	Uncertain $\neq$ definitive fraud finding	Must	UN-01, UN-02
NFR-06	NFR	Reliability	Structured error, no fabricated success	Must	UN-01, UN-06
NFR-07	NFR	Observability	Traceable by request ID/version/latency	Must	UN-03, UN-06
NFR-08	NFR	Compatibility	Dashboard/API same format policy	Should	UN-05
NFR-09	NFR	Maintainability	Model+preprocessing+risk config versioned together	Should	Execution baseline
NFR-10	NFR	Deployability	Containers from README	Must	UN-05
NFR-11	NFR	Maintainability	Automated tests in CI	Should	DOC-08
PC-01	PC	Process	Dataset audit before training	Must	DOC-07
PC-02	PC	Process	Group-aware split or documented limitation	Must	DOC-03
PC-03	PC	Process	Tune on FoR-2sec only	Must	DOC-03
PC-04	PC	Process	Freeze before external test	Must	DOC-03
PC-05	PC	Process	DEEP-VOICE evaluated exactly once	Must	DOC-03

ID	Type	Category	Candidate (short)	Priority	Source
PC-06	PC	Process	Required metric set; no t-DCF	Must	DOC-04
PC-07	PC	Process	Reproducible runs	Must	DOC-07
PC-08	PC	Process	CNN is the single required model	Should	DOC-06
PC-09	PC	Process	Exactly two stretch goals	Should	DOC-06
WH-01	FR	Scope exclusion	Real-time / streaming detection	Won't	DOC-01
WH-02	FR	Scope exclusion	Adversarial-evasion defense	Won't	DOC-01
WH-03	FR	Scope exclusion	Production biometric authenticator claim	Won't	DOC-01
WH-04	FR	Scope exclusion	Second external dataset / alt. model families	Won't	DOC-06

7.3 Validation & Review Log

Validation checklist (evidence-based)

Status	Check	Evidence / Result
PASS	No dataset-method contradiction	The merged document uses FoR-2sec for development and DEEP-VOICE only as the frozen external test throughout; no requirement references ASVspoof, WaveFake, or t-DCF.
PASS	No duplicate runtime requirements	FR-01 is dashboard submission; FR-02 is API submission; each has its own use case (UC-01, UC-02) and shared logic is factored into UC-03/UC-04 rather than duplicated.
PASS	Every final runtime requirement is testable	Each FR/NFR has an acceptance criterion in Sections 4–5; PC-xx items are process rules with their own evidence trail in Section 6.
PASS	Terminology is consistent	Fraud Prevention Analyst, API Client, Prediction, Confidence, Risk level, and Uncertain are defined once (Section 1.3) and used consistently across Sections 2–7.
PASS	Runtime vs. process separation	Dataset audit, split, freeze, tuning, and metric rules are labeled PC-xx (Section 5.7), not runtime NFRs.
PASS	No false-success behavior	Invalid input and inference failures return structured errors (NFR-06); uncertain remains a valid normal result, not an error.
PASS	Decision-support framing preserved	NFR-05 requires all outputs to be framed as decision support; no wording implies a definitive or production-grade biometric finding.
PASS	No diagram-only requirements	Every FR maps to a full detailed use case (UC-01 through UC-10) in Section 3.3; none rely on a box that appears only in Figure 1.
OPEN	Parameterized operational values not yet baselined	L_MAX, MAX_UPLOAD_MB, MAX_AUDIO_SECONDS, RETENTION_HOURS, SUPPORTED_AUDIO_FORMATS, and the risk-mapping thresholds are tracked to closure in Appendix 7.4.

Status	Check	Evidence / Result
OPEN	Independent verification of any external dataset statistics	Any class-count or size figures for DEEP-VOICE should be confirmed directly against the dataset files before being stated in a report; none are asserted in this document for that reason.
OPEN	In-class instructor sign-off	This document incorporates all review rounds completed so far (below); live checkpoint sign-offs remain to be recorded in Section 7.5.

Round 1 — Internal pre-review (merging early drafts)

Issue Found	Our Fix
An early draft still used the old ASVspoof → WaveFake plan with t-DCF.	Aligned everything with the current proposal: FoR-2sec → DEEP-VOICE, no t-DCF.
The classical baseline was missing from evaluation.	Added FR-12 (baseline vs. CNN, same harness).
“Accuracy shall not degrade beyond an agreed margin” was not testable.	Replaced by an explicit gap-reporting requirement (now FR-16).
The rule that DEEP-VOICE is never used for tuning was not a requirement anywhere.	Added the PC-03–PC-05 freeze/one-time-use rules.

Round 2 — In-class peer review (received on the first Lab 3 draft)

Issue Found	Our Fix
Project mismatch: first draft used ASVspoof/WaveFake and t-DCF.	Replaced all methodology references with FoR-2sec → DEEP-VOICE; t-DCF excluded everywhere.
No requirement protected the one-time frozen DEEP-VOICE evaluation.	Added PC-03 to PC-05 and carried them into the SRS constraints section.
Team job titles were treated as stakeholders instead of real personas.	Replaced with Analyst, Customer, Compliance, Operations Manager, API Integration Owner, and Operations/Security Administrator personas.
Elicitation notes looked like rewritten plan text, not raw stakeholder input.	Rewrote as short role-play raw needs/quotes; separated document analysis from interviews.
No performance, security, or privacy requirements existed.	Added measurable, parameterized performance, security, privacy, reliability, compatibility, observability, and maintainability NFRs.
Risk level, confidence, and preprocessing were undefined terms.	Added a fixed-terminology section (now Section 1.3); risk levels are LOW/MEDIUM/HIGH via a versioned mapping policy.
No invalid-input behavior was specified.	Added FR-03 and exception flows in UC-01/UC-02; NFR-06 defines structured failure behavior.
Research workflow rules were counted as runtime NFRs.	Moved dataset, tuning, freeze, metric, and reproducibility rules into PC-xx constraints (Section 5.7).

Issue Found	Our Fix
Use case UML had chained includes and unlabeled use-case links; weak alternate flows.	Redrew the diagram with direct, non-chained includes only (Validate & Analyze Audio, Record Analysis Metadata); added real error flows.
An earlier validation pass claimed consistency without evidence.	Replaced blanket claims with the evidence-based PASS/OPEN checklist above.

Round 3 — Expert review (*independent structural review*)

Issue Found	Our Fix
Traceability cited specific Proposal sections but the elicitation log had only one catch-all “DOC” entry.	Split document-analysis notes into DOC-01–DOC-08, one per cited section, so every Section 6 citation resolves to a specific elicitation-log line.
Several requirements traced only to boxes on the use case diagram with no detailed description.	Every FR now has a full detailed use case in Section 3.3; research-evaluation use cases (UC-07–UC-09) were added specifically to close this gap.
A use case served two actors with one undifferentiated step.	Split dashboard and API submission into separate use cases (UC-01, UC-02), each with a single primary actor, sharing common logic through the included UC-03/UC-04 rather than an internal branch.
A non-functional requirement existed only to restrict an optional feature and carried an awkward conditional priority.	Folded the annotation access restriction directly into FR-11's own requirement text and acceptance criteria; no separate conditional NFR.

Round 4 — Team merge (*this document*)

Both members independently revised the Lab 3 submission after peer review. This document reconciles the two into one, keeping the stronger choice on each point of difference and recording why.

Point of Difference	Resolution
One draft used a single branching use case for submission (dashboard/API); the other split it into two use cases sharing an included validation/analysis step.	Adopted the split model (UC-01, UC-02 + shared UC-03) — each use case keeps one primary actor, which is more standard UML practice, while UC-03/UC-04 avoid duplicating shared logic.
One draft mixed dataset/tuning/freeze rules into NFRs; the other separated them into PC-xx constraints.	Adopted the PC-xx separation (Section 5.7) — it keeps runtime quality attributes and research-process discipline from being mistaken for each other, which is the more defensible SRS structure.
One draft fabricated placeholder numeric targets (e.g., a specific latency value, specific DEEP-VOICE class counts); the other used named, tracked-to-closure parameters throughout.	Adopted the named-parameter approach (L_MAX, MAX_UPLOAD_MB, MAX_AUDIO_SECONDS, RETENTION_HOURS, SUPPORTED_AUDIO_FORMATS; Appendix 7.4) and removed the unverified DEEP-VOICE class-count figure entirely rather than restate it unconfirmed.
One draft had 5 actor-facing use cases (including research evaluation and	Merged both sets: research-evaluation use cases (UC-07–UC-09) and the annotation feature (UC-06) from one draft, plus the Operations/Security Administrator actor

Point of Difference	Resolution
reporting); the other had 3 (submission and history only, with an Operations actor and log-review use case not present in the first).	and Review Service Logs (UC-10) from the other — together giving fuller traceability than either draft alone.
Actor naming differed (“Fraud Analyst” / “Integrating System” vs. “Fraud Prevention Analyst” / “API Client”).	Standardized on “Fraud Prevention Analyst” and “API Client” throughout, as the more precise and consistently-used naming.
One draft's validation log used simple “Checked.” statements; the other used an evidence-based PASS/OPEN status per check.	Adopted the PASS/OPEN evidence format (Section 7.3) as the stronger, more defensible validation style.

Peer feedback received — complete in class

Reviewing Team	Issue Found	Our Fix

7.4 Open Parameters & Questions

The source documents do not fix these values. They are recorded here as named parameters, with a closure rule, instead of being invented.

Parameter / Topic	Meaning / Question	Closure Rule / Affects
L_MAX	Maximum target analysis latency used by NFR-01.	TBD by team after the Week 2 compute/API benchmark; freeze before the performance acceptance test.
MAX_UPLOAD_MB	Maximum accepted upload size.	TBD by team before API/dashboard integration tests. Affects FR-03, NFR-08.
MAX_AUDIO_SECONDS	Maximum accepted audio duration for the short-audio prototype (model is trained on 2-second clips — how longer clips are handled needs a rule).	TBD by team before validation tests. Affects FR-03, FR-02, NFR-01.
RETENTION_HOURS	Maximum raw-upload retention period.	TBD with the Compliance Officer; use a short value in automated retention tests. Affects NFR-03.
SUPPORTED_AUDIO_FORMATS	Allowed audio formats for dashboard and API.	TBD and documented once the ingestion path is finalized; the same set must apply to both interfaces (NFR-08).
Risk-level mapping policy	Exact thresholds and versioning for mapping prediction + confidence to LOW/MEDIUM/HIGH.	Must be finalized on FoR-2sec only and frozen before DEEP-VOICE external testing

Parameter / Topic	Meaning / Question	Closure Rule / Affects
		(consistent with PC-03/PC-04). Affects FR-07.
History persistence	Must the analysis history survive service restarts and be shared across analysts?	Affects FR-10; to be decided with the storage/deployment design.
Access control mechanism	What authentication do the dashboard and API require (e.g., API key, analyst login) to support FR-11's access restriction?	Affects FR-11, NFR-02.

7.5 Lab 3 Checkpoints & Instructor Sign-off

Checkpoint	Evidence in this document	Instructor sign-off / date
1 — Stakeholders & elicitation	2.3, 7.1	
2 — Requirements analysis (MoSCoW)	7.2	
3 — Top 3 Must-haves read aloud	FR-05, FR-09, FR-15	
4 — Use case diagram + descriptions	3.2, 3.3	
5 — Peer review and revisions	7.3 (Rounds 1–4)	
6 — Draft SRS, Sections 4–5	4, 5	

7.6 Glossary

Term	Definition
Synthetic / spoofed speech	Audio generated or manipulated (text-to-speech or voice conversion) to imitate a real speaker.
Bonafide	Genuine human speech; the dataset term for “authentic”.
Text-to-speech (TTS)	Generating speech from text. FoR-2sec's synthetic clips are TTS.
Voice conversion	Transforming one person's speech into another's voice. DEEP-VOICE's synthetic clips use RVC.
Cross-dataset / external generalization	How well a model keeps its performance on a dataset it was not trained on.
Frozen model	The final model whose weights and decision thresholds may no longer change.
Group-aware split	A data split in which all clips from one speaker or source recording go to the same split.
Risk-mapping policy	The versioned logic that turns a prediction and confidence score into a LOW/MEDIUM/HIGH operational risk level.
Inference preprocessing profile	The documented, versioned preprocessing (resampling, trimming, feature extraction) applied identically to every submitted clip.

Term	Definition
Log-mel spectrogram	Time-frequency representation of audio on the mel scale; the CNN's input.
t-DCF	ASVspoof-specific cost metric; noted for reference only and not used in this project.
Decision support	A system role that assists human judgment rather than issuing an automatic or definitive verdict — core framing for all analysis outputs (NFR-05).